GFN1000
In Dialogue with Nature

Aristotle/Lindberg

Core Question

How do we handle what nature tells us?

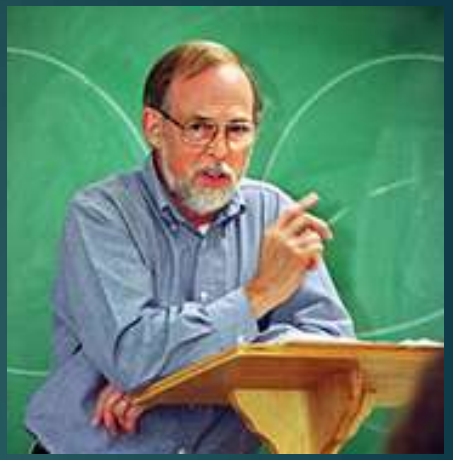


David Lindberg

- American historian of science
- Main focus:

  History of medieval and early modern science, especially physical science and the relationship between religion and science.

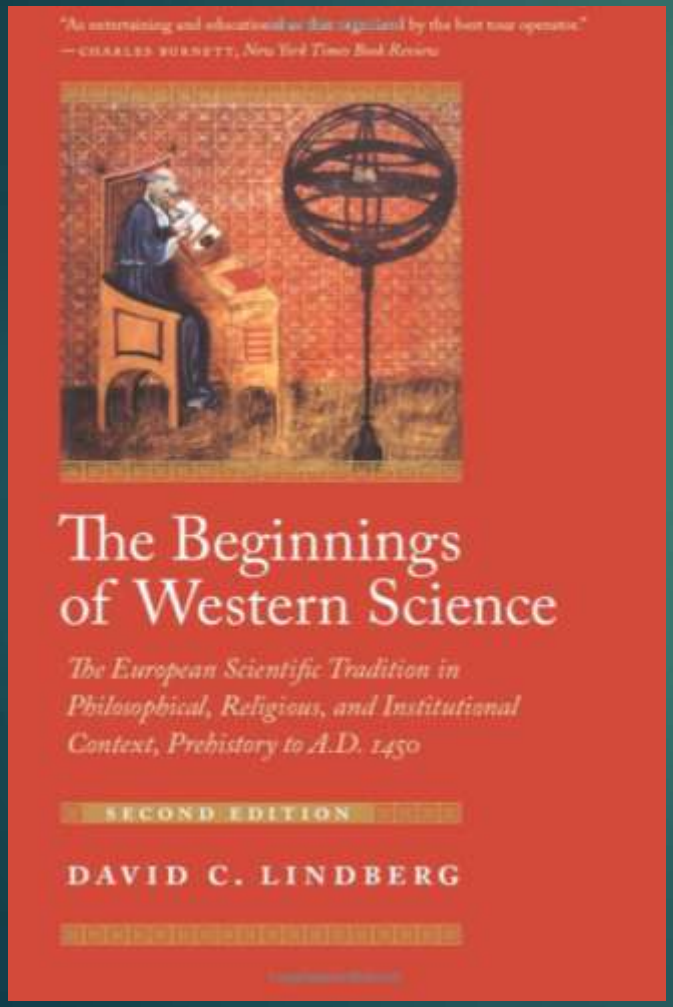
- The recipient of the Sarton Medal (萨顿勋章) in 1999 (the medalists include Joseph Needham in 1968, I. Bernard Cohen in 1974)



David C. Lindberg (1935-2015)

*The George Sarton Medal is the most prestigious award given by the History of Science Society.

The Beginnings of Western Science



- A textbook for the history of science

  Chapter 1: Science Before the Greeks

  Chapter 2: The Greeks and the Cosmos (Text 1b)

  Chapter 3: Aristotle's Philosophy of Nature

  Chapter 4: Hellenistic Natural Philosophy

  ……

  Chapter 11: The Medieval Cosmos

  Chapter 12: The Physics of the Sublunar Region

  ……

Aristotle's profile



Aristotle
(384 BC – 322 BC)

- A natural philosopher (自然哲学家)
- Student of Plato, joined Academy (17 yr), leave there (37 yr)
- Tutored Alexander the Great (亚历山大大帝, ~42 yr)
- Established the school Lyceum (~50 yr)
  - Shifted from Platonism to Empiricism (经验主义)
- Founded the science of Logic (逻辑学)
- The last of the great Greek philosophers, Europe's first great biologist.

"I love Plato, but I love Truth more."

吾爱吾师，吾更爱真理。

bio.COM

Chapter 3: Aristotle's Philosophy of Nature

Metaphysics & Epistemology

Nature and Change
– how change happens and be explained

Cosmology
– Aristotelian universe: sublunar and supralunar region

Motion
– terrestrial and celestial motions

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Aristotelianism *versus* Platonism



`*School of Athens'(*1511) by Raphael

Aristotelianism *versus* Platonism



The only thing is real is the world of ideas or forms.

Aristotle rejects the idea of form as separate reality, he believes the sensible objects make up the real world.

Metaphysics (形而上学) and Epistemology (认识论)

- **Plato**: naturally directed toward the eternal forms, knowable through reason or philosophical reflection.
- **Aristotle's** metaphysics of concrete individuals directed his quest for knowledge toward the material world of individuals, of nature, and of change – a world encountered through the senses.

Acquiring knowledge begins with experience…

Logic

Inductive (归纳) reasoning:

- **start with specific observations and form general conclusions.**

- **“But what we learn by this “inductive” process does not acquire the status of true knowledge until put into deductive form; the end product is a deductive demonstration beginning from universal definitions as premises.”**

Deductive (演绎) reasoning:

a logical approach where you progress from general ideas to specific conclusions.

Inductive (归纳) reasoning:

✓Lili likes 8:30 lessons.

✓Lucy likes 8:30 lessons.

➔All university students like 8:30 lessons.

(The conclusion is possibly false, even if all of the premises are true.)

Deductive (演绎) reasoning:

✓All university students like 8:30 lessons.

✓Lili is a university student.

➔Lili likes 8:30 lessons.

(The conclusion is true if all the premises (前提) are true.)

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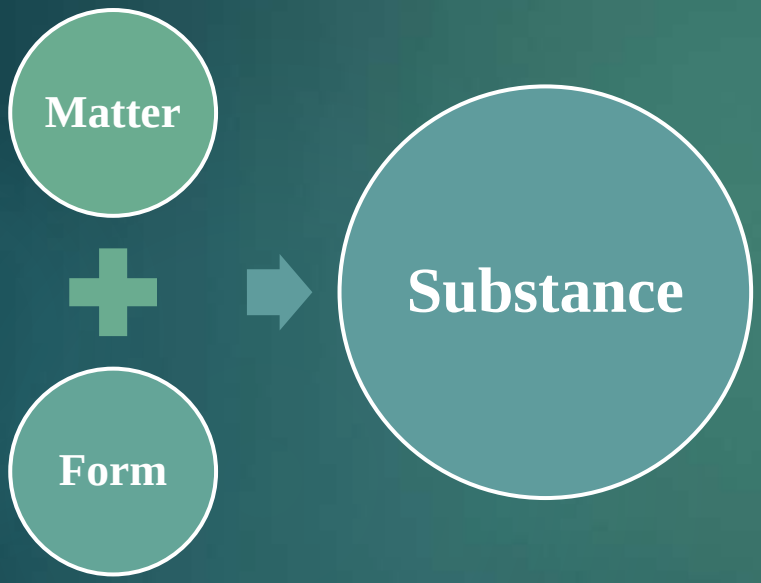
Motion
– terrestrial and celestial motions

How change happens?

- Plato separated world of sense from World of Forms (理型)
  - He restricted change to the imperfect material replica of the world of forms.

- Aristotle rejects the idea of Form as separate reality
  - He explains change with form (形式) and matter (质料).

Form and Matter



- Individual things (substances) are a combination of **form** and **matter.**
- **Matter** serves as the subject for the form.
- **Form** consists the properties that make the substance what it is.
- **Matter and form** are always combined (but intellectually distinct).

Explaining Change:

- When an object undergoes change, its form changes while its matter remains unchanged.

- Change occurs in **substance/subject** moving from a property to its opposite (**pairs of contrary qualities**).

  Such as dry becomes wet, or cold becomes hot...

Parmenides (515BC-c.450BC)

About reality

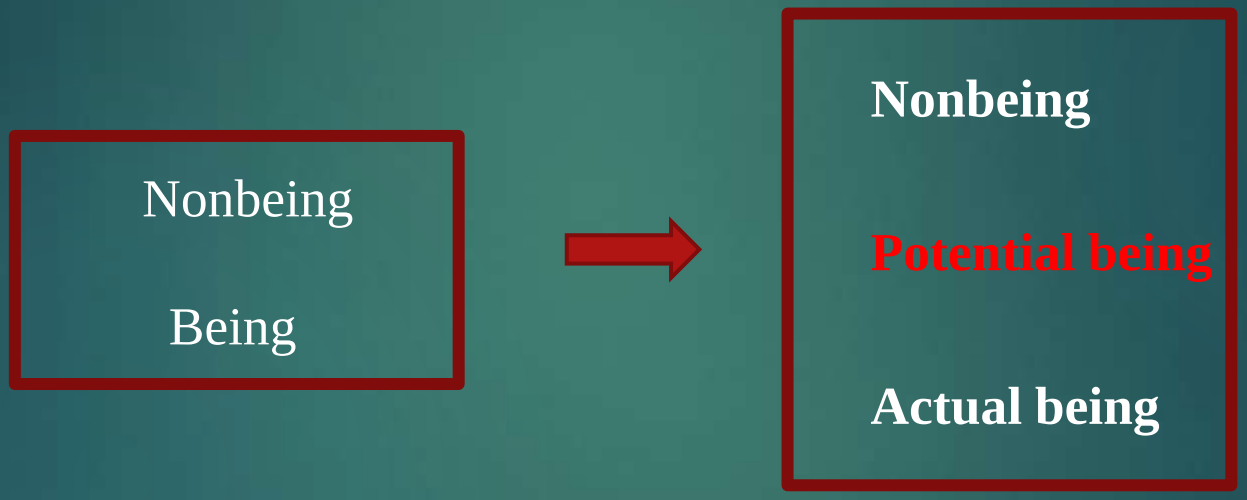
- reality is one, change is impossible, existence is timeless, uniform, necessary, and unchanging

- “nothing comes from nothing” (没有任何事物来自虚无)

- sensory faculties lead to conceptions which are false and deceitful

Aristotle's resolution: potential being

- One more state is associated with being:



- Every change in nature is the result from potential being to actual being.

Potentiality- Actuality

- An acorn is potentially (but not actually) an oak tree.





Why an acorn stops remaining in its original status?

Theory of nature and causation

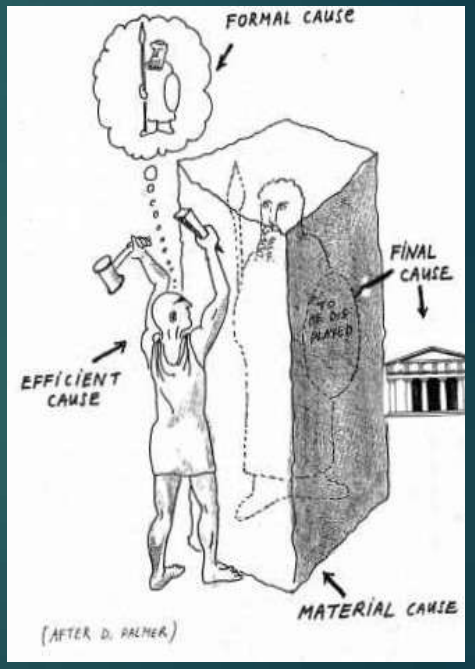
- Why an acorn stops remaining in its original status?

  -The nature of an acorn is to do so.

- Aristotle believes that every natural object has a nature or an attribute that makes the object behave in its fashion.

The four causes

Material cause	What is it made of? Ex. The statue is made of marble (大理石)
Formal cause	What is its shape or structure? Ex. The statue is in the shape of man
Efficient cause	What is the agency that brings about the change? Ex. The sculptor (雕塑家)
Final cause	What is its purpose served by the change? Ex. To be displayed; to be beautiful.



Aristotle's teleological (目的论的) *cosmos*

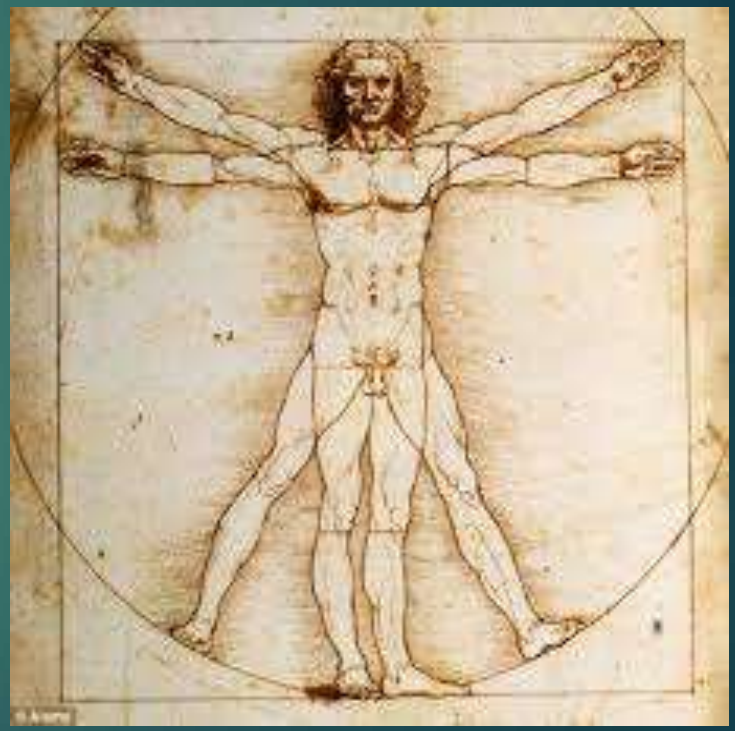
- Final causes are very significant to Aristotle's philosophy.
- The entire *cosmos* has a *telos(*目的*)*.
- Teleological explanations were dominant until Charles Darwin.



What are the *causes* of human being?

- **Material cause**:

  Flesh, bones… (oxygen, carbon, hydrogen, nitrogen, calcium, and phosphorus, to be more exact)

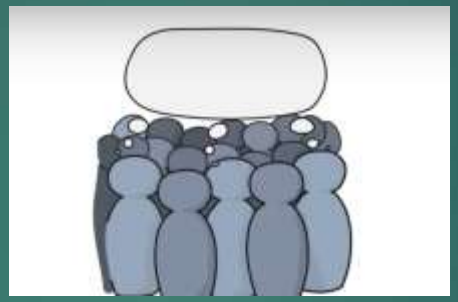


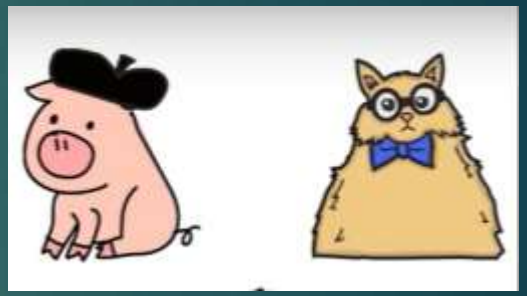
- **Efficient cause**:
  parents (human being)
- **Formal cause:**

  form (human being)
- **Final cause**:
  What is the purpose to be human?

What brings happiness?













What brings happiness?

- Happiness is a certain kind of activity of the soul in accordance with complete virtue.
  - Intellectual virtue
  - Moral virtue



Chapter 3: Aristotle's Philosophy of Nature

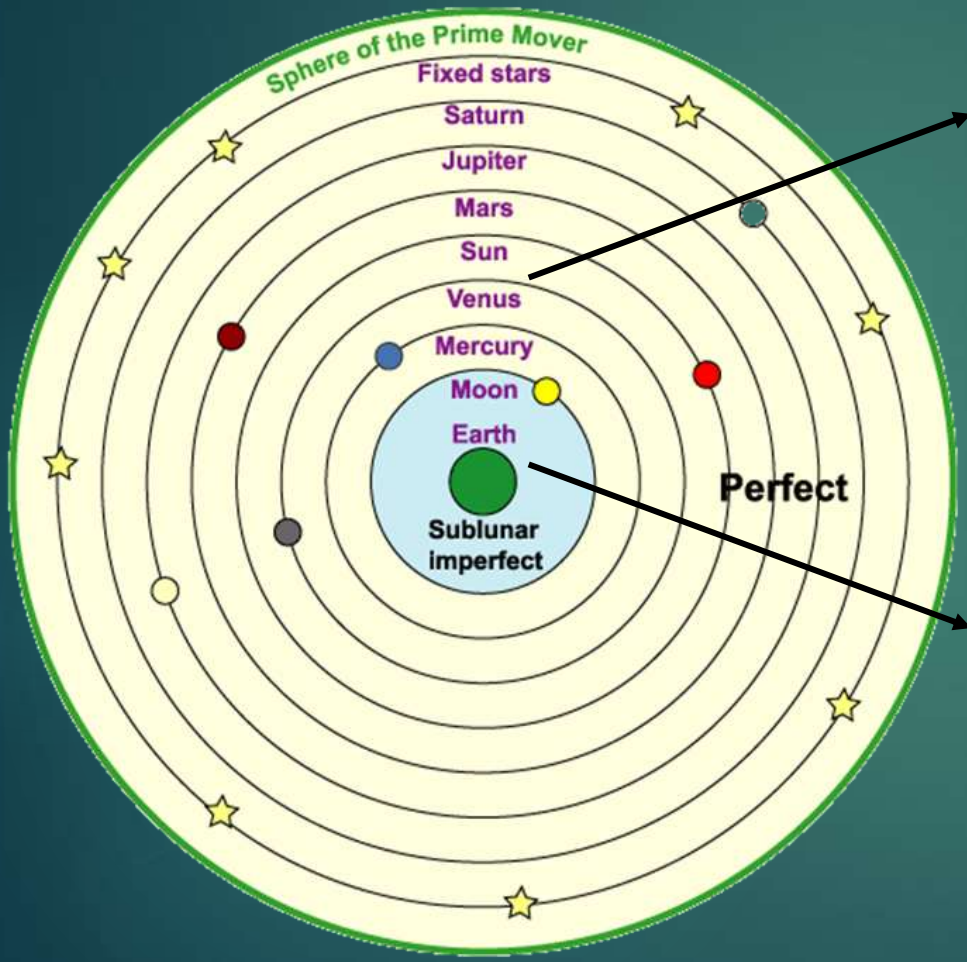
Metaphysics & Epistemology

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The Aristotelian cosmos (宇宙)

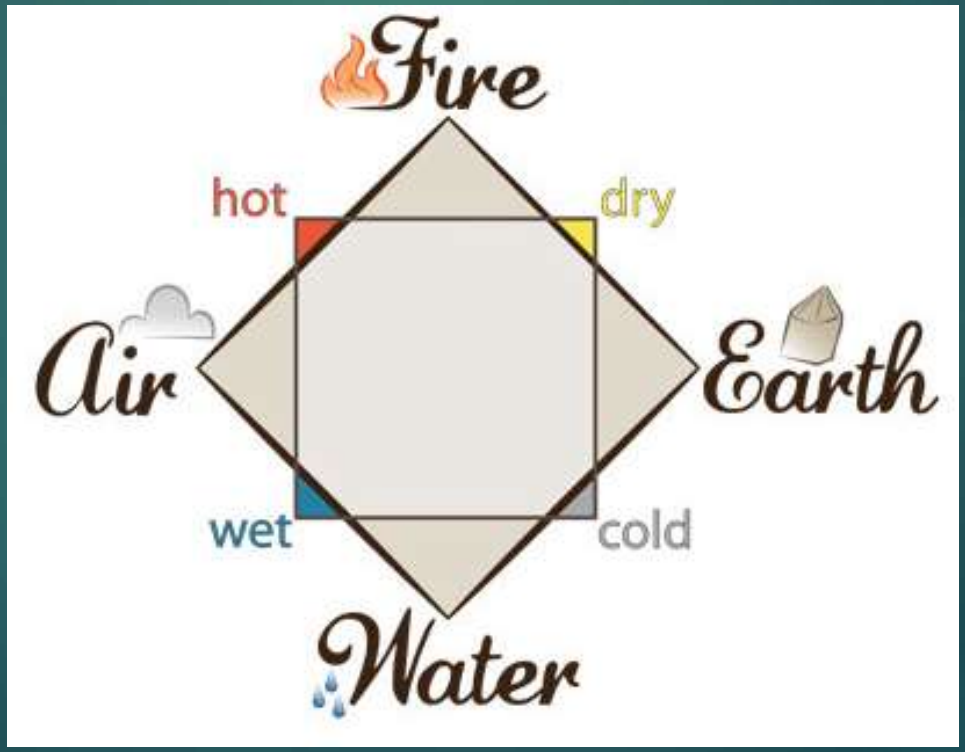


Celestial/ supralunar region:
Perfect;
Eternally unchanging;
Made of element of aether (以太).

Terrestrial/ sublunar region:
Imperfect;
Changing;
Made of four elements
(fire, air, water, earth).

Aristotelian elements and qualities

- Aristotle accepted the four elements originally proposed by Empedocles —earth, water, air, and fire.
- Aristotle choose sensible qualities as the ultimate building blocks.



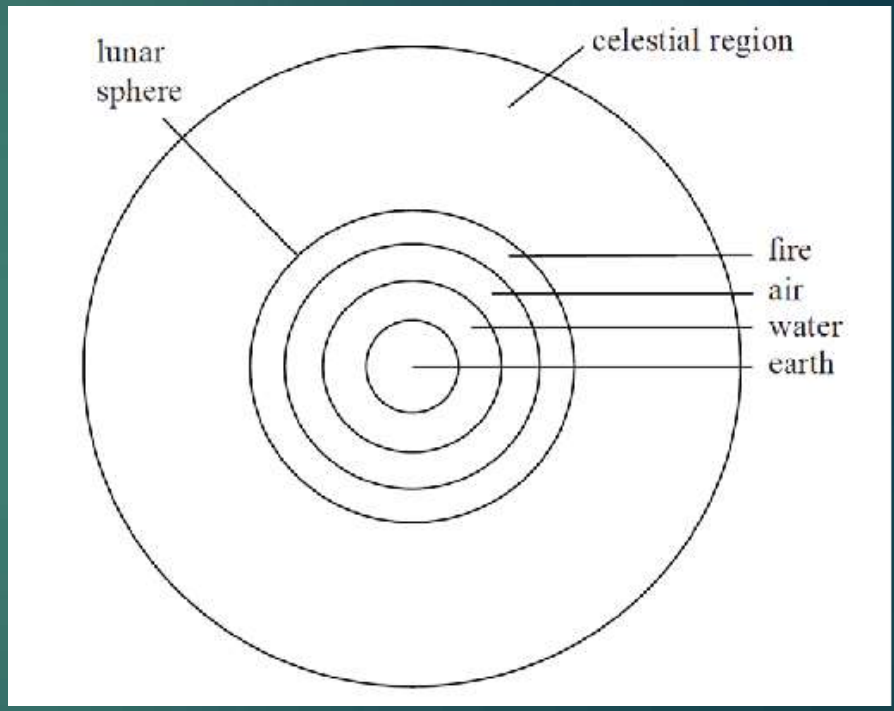
The Arestotelian *cosmos*

Each of the elements is also heavy or light, they either go straight up or straight down.



e.g. fire goes up,

earth goes down.



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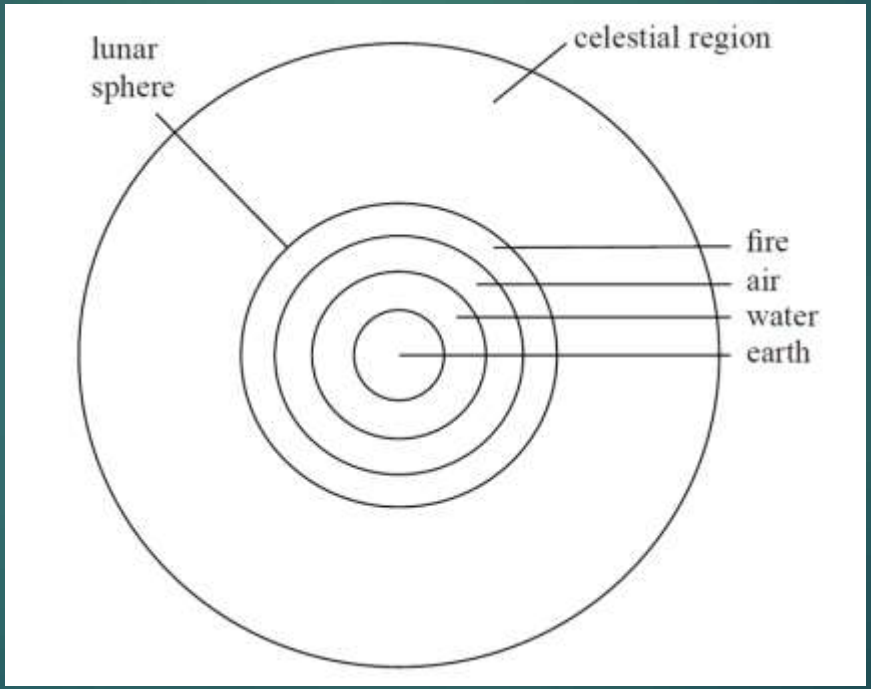
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– terrestrial and celestial motions

Theory of motion

- Motion is never spontaneous; there is no motion without a mover.
- “Natural” motion & “Forced” or “violent” motion.

Natural motion

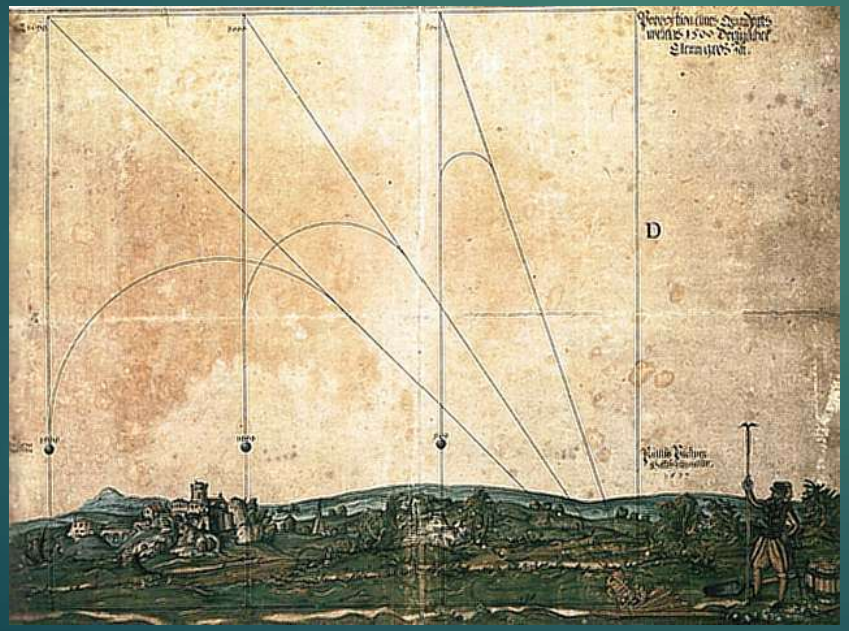
-motion toward the natural place of the moving body
-moving body ceases when it reaches its natural place

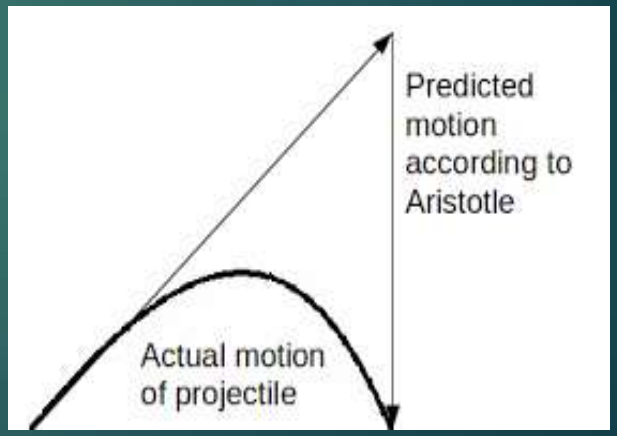


Mover: the nature of the body.

Unnatural or "Violent" Motion

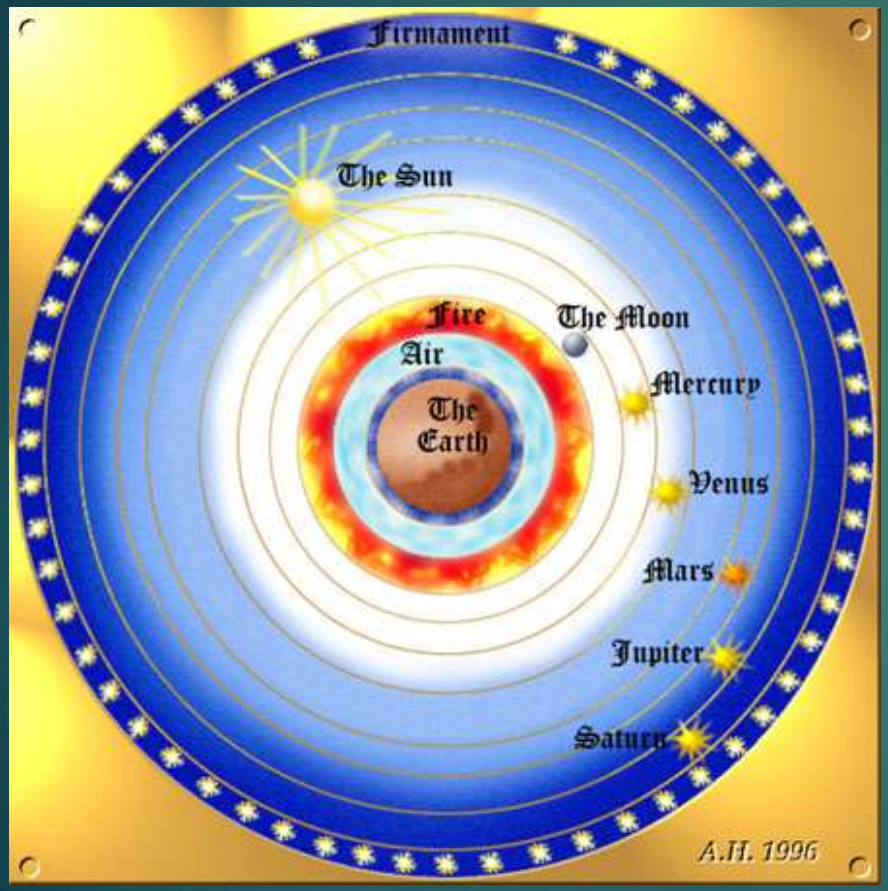
- Motion in any other direction occurs only under an outside force. Mover: external force.
- As soon as the force ceases the motion ceases.





Celestial (天空的) Motion

-continuous uniform circular motion



Mover: Prime Mover

Aristotle identified the unmoved mover for the planetary spheres as the "Prime Mover."

- a living deity representing the highest good, wholly actualized, totally absorbed in self-contemplation, nonspatial, separated from the spheres it (or he or she) moves, and not at all like the traditional anthropomorphic Greek gods.

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Aristotle's Philosophy of Nature

Perhaps the most important point to be made about final cause is its clear 22
illustration of the role of purpose (the more technical term is "teleology") in Aristotle's universe. The world of Aristotle is not the inert, mechanistic world of the atomists, in which the individual atom pursues its own course mindless of all others. Aristotle's world is not a world of chance and coincidence, but an orderly, organized world, a world of purpose, in which things develop toward ends determined by their natures. It would be unfair and pointless to judge Aristotle's success by the degree to which he anticipated modern science (as though his goal was to answer our questions, rather than his own); it is nonetheless worth noting that the emphasis on functional explanation to which Aristotle's teleology leads would prove to be of profound significance for all of the sciences and remains to this day a dominant mode of explanation within the biological sciences.

Chapter 12:
The Physics of the Sublunar Region

Medieval physics was not a primitive version of modern physics and cannot be 1
legitimately judged by comparison with its modern namesake. Certainly there is
overlap between the two, but medieval physics, deeply engaged with Aristotelian
metaphysics and natural philosophy devoted much more attention to fundamental
issues that we would classify as "metaphysical" or "philosophical"; concerning the
fundamental stuff of the universe, the elements and their constituents, the sources
of motion and change, and the like.

The medieval natural philosopher (as I will generally refer to him) took his 2
starting point from the text of Aristotle's *Physics* and other works, and devoted
his career to clarification of ambiguities, disputation about difficult or contentious
portions of the text, and original application or extension of Aristotelian principles.
But he was emphatically *not* a slave to the Aristotelian text, as a widespread myth
would have it; rather, he was typically a gifted reader and interpreter of the texts of
Aristotle and his commentators (including critics), eager to display his logical and

The Physics of the Sublunar Region

(The Middle Ages)
Understanding of
Motion

kinematics – description of motion: emergence of mathematical and geometrical descriptions

dynamics – causes of motion: identify mover; quantify the relationship between force and velocity.

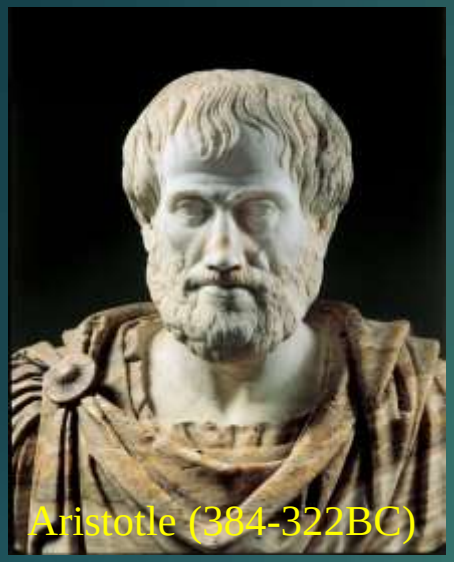
The Physics of the Sublunar Region

- Mathematical Description of Motion (kinematics)
  - Merton scholars
    - Made explicit distinction between kinematics and dynamics
    - Developed a conceptual framework: “velocity”, “instantaneous velocity”, …
    - Made distinction between uniform and nonuniform motion
    - Developed a variety of kinematic theorems
  - Intensity and quantity of a “quality” or “form”
  - Nicole Oresme’s geometrical proof of the Merton rule

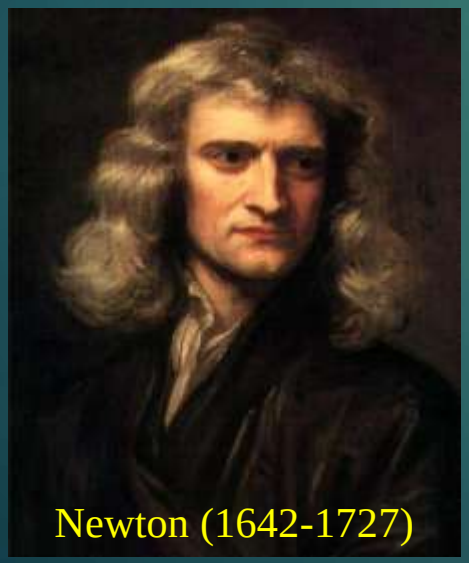
- The Dynamics of Local Motion

  E.g. How to explain projectile motion

The dynamics of local motion



Aristotle (384-322BC)



Newton (1642-1727)

6th John Philoponus

10-12th Avicenna & Averroses

13th Thomas Aquinas

14th Franciscus de Marchia & John Buridan

16th-17th Galileo Galilei

15th-16th Nicolaus Copernicus

16th-17th Johannes Kepler

Reading assignment

- Read Text 2

Lindberg “The Beginnings of Western Science”

Notification

Quiz（Text 1 & 2） in next tutorial !
Be punctual !

Bibliography

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